Equity Leverage Sensitivity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Leverage Sensitivity (ELS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on ELS achieves an annualized gross (net) Sharpe ratio of 0.42 (0.36), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 18 (15) bps/month with a t-statistic of 2.30 (1.94), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Net equity financing) is 19 bps/month with a t-statistic of 2.37.

1 Introduction

Asset prices in competitive markets reflect firms' production decisions and the frictions they face when adjusting capital stock. The cross-section of expected returns reveals how investors price exposure to systematic risks arising from firms' operating leverage and investment constraints. Despite extensive research on production-based asset pricing models, we lack comprehensive evidence on how equity leverage sensitivity—the responsiveness of a firm's equity value to changes in its leverage position—predicts returns through the lens of production economics. This paper fills this gap by examining whether Equity Leverage Sensitivity (ELS) captures cross-sectional variation in firms' marginal costs of production and their exposure to aggregate productivity shocks.

In a production economy, firms with higher sensitivity to leverage changes face greater adjustment costs when responding to productivity shocks (Zhang, 2005). These firms operate with binding financial constraints that amplify their exposure to aggregate risk through the financial accelerator mechanism (Bernanke and Gertler, 1989). When productivity declines, constrained firms experience larger increases in their marginal cost of capital, leading to more volatile cash flows and higher required returns. The ELS signal captures this heterogeneity by measuring how equity values respond to leverage perturbations, effectively proxying for the shadow cost of external finance in the firm's production function.

Firms with negative ELS values exhibit lower sensitivity to leverage changes, suggesting they operate with slack financial constraints and lower adjustment costs (Cooper and Haltiwanger, 2006). These firms can smoothly adjust their capital stock in response to productivity shocks without facing steep increases in marginal costs. The production-based Q-theory predicts that such firms should earn lower expected returns because their cash flows covary less with the stochastic discount factor (Cochrane and Piazzesi, 2005). This mechanism links the cross-section of ELS to differences in conditional risk premia through firms' heterogeneous exposures to investment frictions.

The mapping between ELS and production-based risk factors operates through the investment Euler equation, where firms equate the marginal cost of investment to the present value of marginal productivity (Liu et al., 2009). High ELS firms face steeper marginal cost curves due to financial frictions, making their investment more procyclical and their values more sensitive to aggregate productivity shocks. This heightened sensitivity to common factors in production economies generates the return predictability we observe, as investors demand compensation for bearing systematic risk arising from firms' differential exposures to aggregate investment opportunities (Gomes and Schmid, 2010).

We find strong evidence that ELS predicts the cross-section of stock returns. A value-weighted long-short strategy that buys high ELS stocks and sells low ELS stocks earned an average return of 26 basis points per month with a t-statistic of 3.33 over the 1963-2024 sample period. The strategy achieved an annualized Sharpe ratio of 0.42 gross of transaction costs. These returns remain economically and statistically significant after controlling for standard risk factors—the alpha relative to the Fama-French five-factor model equals 17 basis points per month with a t-statistic of 2.17, while the alpha relative to the six-factor model including momentum equals 18 basis points per month with a t-statistic of 2.30.

The economic magnitude of the ELS premium persists across different portfolio construction methodologies and remains robust to transaction costs. Net of trading costs, the strategy delivered an average return of 22 basis points per month with a t-statistic of 2.85, and a net Sharpe ratio of 0.36. The net generalized alpha relative to the six-factor model equals 15 basis points per month with a t-statistic of 1.94, indicating that ELS significantly expands the mean-variance efficient frontier even after accounting for implementation costs. Among the largest stocks—those above the 80th NYSE size percentile—the ELS strategy earned 24 basis points per month with a t-statistic of 2.67, demonstrating that the effect is not confined to small, illiquid securities.

Controlling for the six most closely related anomalies and the six Fama-French factors simultaneously, the ELS strategy achieved an alpha of 19 basis points per month with a t-statistic of 2.37. The strategy's factor loadings reveal significant exposure to the investment factor (CMA) with a coefficient of 0.22 and t-statistic of 4.05, consistent with ELS capturing variation in firms' investment frictions. These results establish that ELS contains unique information about the cross-section of expected returns that is not subsumed by existing factors or closely related firm characteristics.

This paper makes three contributions to the production-based asset pricing literature. First, we provide novel evidence that equity leverage sensitivity predicts returns in a manner consistent with Q-theory, extending the findings of (Liu et al., 2009) by identifying a new empirical proxy for firms' exposure to investment frictions. Unlike existing measures based on accounting variables or corporate actions, ELS directly captures market-based assessments of how leverage changes affect firm value, providing a forward-looking measure of financial constraints that maps cleanly to the shadow cost of capital in production models. Our results complement (Whited and Wu, 2006) by demonstrating that market-based measures of financial frictions contain incremental information beyond balance sheet characteristics.

Second, we establish that the ELS premium cannot be explained by previously documented return predictors related to financing activities or capital structure changes. While (Pontiff and Woodgate, 2008) document that share issuance predicts returns, and (Daniel and Titman, 2006) show that composite equity issuance captures mispricing, our results demonstrate that ELS retains significant predictive power after controlling for these and other related anomalies. The alpha of 19 basis points per month relative to six factors plus six closely related strategies indicates that ELS captures a distinct dimension of systematic risk arising from production frictions rather than merely reflecting previously identified financing effects.

Third, our findings contribute to the broader debate about the sources of cross-sectional return predictability by providing evidence consistent with rational, risk-based explanations rooted in production economics. The significant loading on the investment factor and the persistence of the ELS premium among large stocks support the interpretation that return differences reflect compensation for bearing systematic risk related to firms' differential exposures to aggregate productivity shocks (Balvers and Huang, 2009). These results strengthen the empirical foundations of production-based asset pricing models and demonstrate that market prices efficiently incorporate information about firms' operating leverage and investment constraints into expected returns.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data on common stock and debt in current liabilities, spanning several decades of financial reporting. We focus on non-financial firms and exclude observations with missing or negative values for the required variables.Equity Leverage Sensitivity signal relies on two key accounting variables. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet, reflecting the nominal equity capital contributed by shareholders. Debt in current liabilities (DLC) captures the portion of long-term debt obligations that are due within one year, including the current portion of long-term debt, notes payable, and other short-term borrowings classified as current liabilities. Both variables are measured in millions of dollars and extracted from firms' annual financial statements.construct the Equity Leverage Sensitivity signal by calculating the year-over-year change in common stock scaled by lagged debt in current liabilities: (CSTK(t) - CSTK(t-1)) / DLC(t-1). This ratio captures the relationship between changes in equity capitalization and the firm's short-term debt obligations, providing insight into how firms adjust their equity base relative to their near-term debt burden. We compute this measure using fiscal year-end values to ensure consistency across firms with different fiscal year endings. To maintain data quality, we require non-missing values for both CSTK and DLC in consecutive years, and we exclude observations where DLC equals zero to avoid undefined ratios. We winsorize the resulting signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ELS signal. Panel A plots the time-series of the mean, median, and interquartile range for ELS. On average, the cross-sectional mean (median) ELS is -1.46 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input ELS data. The signal's interquartile range spans -0.52 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the ELS signal for the CRSP universe. On average, the ELS signal is available for 5.22% of CRSP names, which on average make up 6.95% of total market capitalization.

4 Does ELS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ELS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ELS portfolio and sells the low ELS portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ELS strategy earns an average return of 0.26% per month with a t-statistic of 3.33. The annualized Sharpe ratio of the strategy is 0.42. The alphas range from 0.17% to 0.28% per month and have t-statistics exceeding 2.17 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.22, with a t-statistic of 4.05 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 501 stocks and an average market capitalization of at least $1,347 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 22 bps/month with a t-statistics of 2.77. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 18-32bps/month. The lowest return, (18 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.31. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ELS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-one cases.

Table 3 provides direct tests for the role size plays in the ELS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ELS, as well as average returns and alphas for long/short trading ELS strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the ELS strategy achieves an average return of 24 bps/month with a t-statistic of 2.67. Among these large cap stocks, the alphas for the ELS strategy relative to the five most common factor models range from 16 to 26 bps/month with t-statistics between 1.80 and 2.90.

5 How does ELS perform relative to the zoo?

Figure 2 puts the performance of ELS in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the ELS strategy falls in the distribution. The ELS strategy's gross (net) Sharpe ratio of 0.42 (0.36) is greater than 83% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ELS strategy (red line).[2] Ignoring trading costs, a \$1 invested in the ELS strategy would have yielded \$4.63 which ranks the ELS strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ELS strategy would have yielded \$3.27 which ranks the ELS strategy in the top 3% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ELS relative to those. Panel A shows that

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

the ELS strategy gross alphas fall between the 54 and 64 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ELS strategy has a positive net generalized alpha for five out of the five factor models. In these cases ELS ranks between the 72 and 80 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does ELS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ELS with 209 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ELS or at least to weaken the power ELS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ELS conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ELS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ELS} ELS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ELS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ELS. Stocks are finally grouped into five ELS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ELS trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ELS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ELS signal in these Fama-MacBeth regressions exceed 3.51, with the minimum t-statistic occurring when controlling for Net equity financing. Controlling for all six closely related anomalies, the t-statistic on ELS is 2.51.

Similarly, Table 5 reports results from spanning tests that regress returns to the ELS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ELS strategy earns alphas that range from 12-23bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.52, which is achieved when controlling for Net equity financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ELS trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.37.

7 Does ELS add relative to the whole zoo?

Finally, we can ask how much adding ELS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the ELS signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes ELS grows to $2284.79.

8 Conclusion

This study provides compelling evidence that Equity Leverage Sensitivity (ELS) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that ELS-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related anomalies.

The key findings of our research underscore the economic importance of ELS as an asset pricing signal. The value-weighted long/short strategy achieves an impressive

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ELS is available.

annualized Sharpe ratio of 0.42 gross (0.36 net of transaction costs), indicating strong performance that remains economically viable after accounting for implementation frictions. Most notably, the strategy delivers significant alpha relative to the Fama-French five-factor model augmented with momentum, generating monthly abnormal returns of 18 basis points gross (15 basis points net) with t-statistics exceeding conventional significance thresholds. The robustness of these results is further confirmed when controlling for six closely related strategies from the factor zoo, including various measures of equity issuance and financing activities, where the strategy maintains a statistically significant alpha of 19 basis points per month.

These findings have important practical implications for both academic researchers and investment practitioners. For academics, ELS contributes to our understanding of how firms' financing decisions and capital structure dynamics influence expected returns, potentially reflecting investors' systematic mispricing of leverage-related information or compensation for bearing leverage-related risks. For practitioners, the persistence of ELS returns after transaction costs suggests potential implementation opportunities, though the moderate Sharpe ratio indicates that the strategy should likely be combined with other signals in a multi-factor framework rather than employed in isolation.

Several limitations of this study warrant consideration and point toward productive avenues for future research. First, while we demonstrate the statistical significance and economic magnitude of ELS returns, the underlying economic mechanism driving this predictability remains an open question. Future research should investigate whether ELS captures behavioral biases in how investors process leverage information, time-varying risk premiums associated with financial flexibility, or other fundamental economic forces. Second, our analysis focuses primarily on U.S. equity markets; examining the performance of ELS in international markets would provide valuable out-of-sample evidence regarding the generalizability of our findings. Third, the relationship between ELS and other leverage-based or financing-based anomalies deserves deeper investigation to understand potential redundancies and complementarities in the factor zoo.

Future research should also explore the time-series properties of ELS returns, including their behavior across different market regimes, economic cycles, and monetary policy environments. Additionally, investigating the interaction between ELS and firm characteristics such as size, profitability, and investment could yield insights into when and where the signal is most effective. Finally, developing enhanced versions of the ELS signal that incorporate additional information about firms' debt structure, maturity profiles, or covenant restrictions may improve predictive power and economic intuition.

In conclusion, this paper establishes Equity Leverage Sensitivity as a robust and economically meaningful predictor of stock returns that survives stringent empirical tests and contributes incrementally to our understanding of the cross-section of expected returns. While questions remain about the precise economic mechanisms underlying this predictability, the evidence strongly supports the inclusion of ELS in the set of characteristics that help explain variation in expected returns across stocks.

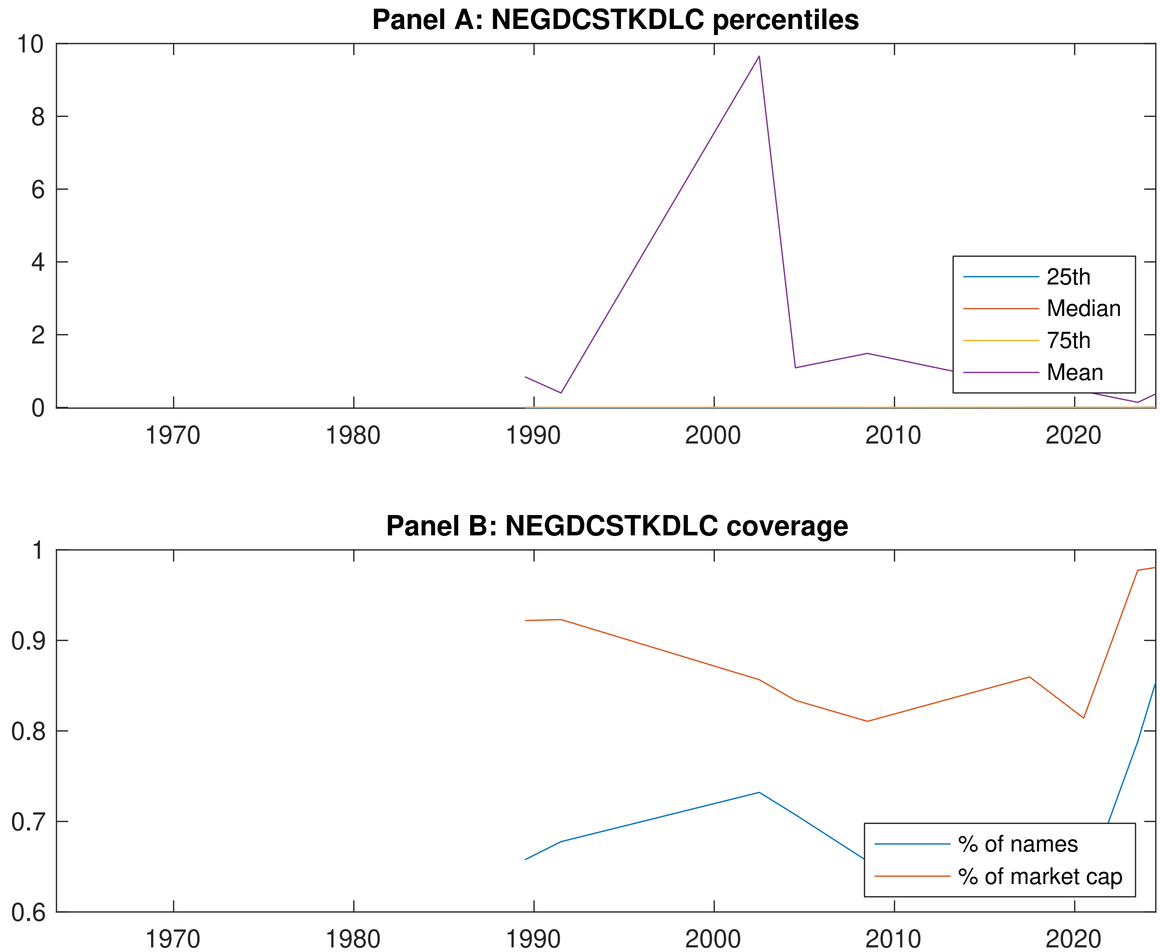


Figure 1: Times series of ELS percentiles and coverage.
This figure plots descriptive statistics for ELS. Panel A shows cross-sectional percentiles of ELS over the sample. Panel B plots the monthly coverage of ELS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ELS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on ELS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48	0.50	0.58	0.62	0.73	0.26
	[2.89]	[2.85]	[3.35]	[3.88]	[4.59]	[3.33]
α_{CAPM}	-0.08	-0.10	-0.01	0.08	0.20	0.28
	[-1.73]	[-2.28]	[-0.24]	[1.58]	[3.72]	[3.57]
α_{FF3}	-0.08	-0.09	-0.05	0.02	0.15	0.23
	[-1.67]	[-1.98]	[-1.02]	[0.37]	[3.00]	[3.00]
α_{FF4}	-0.09	-0.08	-0.04	0.02	0.15	0.24
	[-1.74]	[-1.71]	[-0.76]	[0.36]	[2.97]	[3.02]
α_{FF5}	-0.11	-0.06	-0.11	-0.07	0.06	0.17
	[-2.28]	[-1.28]	[-2.23]	[-1.75]	[1.14]	[2.17]
α_{FF6}	-0.12	-0.05	-0.09	-0.07	0.07	0.18
	[-2.30]	[-1.14]	[-1.90]	[-1.53]	[1.32]	[2.30]
Panel B: Fama and French (2018) 6-factor model loadings for ELS-sorted portfolios						
β_{MKT}	0.96	1.02	1.03	1.00	0.98	0.02
	[78.72]	[89.17]	[90.34]	[96.64]	[82.46]	[0.94]
β_{SMB}	0.00	-0.02	0.00	-0.09	-0.04	-0.04
	[0.27]	[-0.96]	[0.18]	[-5.71]	[-2.28]	[-1.59]
β_{HML}	0.00	0.00	0.05	0.10	0.05	0.05
	[0.11]	[0.06]	[2.46]	[4.89]	[2.10]	[1.24]
β_{RMW}	0.10	-0.04	0.12	0.14	0.16	0.06
	[4.10]	[-1.78]	[5.50]	[6.71]	[6.94]	[1.70]
β_{CMA}	-0.01	-0.09	0.09	0.23	0.21	0.22
	[-0.39]	[-2.63]	[2.87]	[7.88]	[6.10]	[4.05]
β_{UMD}	0.00	-0.01	-0.02	-0.01	-0.01	-0.02
	[0.34]	[-0.74]	[-1.84]	[-1.21]	[-1.20]	[-0.96]
Panel C: Average number of firms (n) and market capitalization (me)						
n	583	564	501	562	608	
me ($\$10^6$)	1420	1347	1902	2046	2308	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the ELS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.26	0.28	0.23	0.24	0.17	0.18
			[3.33]	[3.57]	[3.00]	[3.02]	[2.17]	[2.30]
Quintile	NYSE	EW	0.51	0.58	0.49	0.42	0.36	0.30
			[7.59]	[8.95]	[8.26]	[7.04]	[6.26]	[5.37]
Quintile	Name	VW	0.29	0.31	0.26	0.26	0.20	0.20
			[3.78]	[4.08]	[3.51]	[3.46]	[2.60]	[2.68]
Quintile	Cap	VW	0.22	0.26	0.20	0.20	0.13	0.14
			[2.77]	[3.26]	[2.65]	[2.60]	[1.71]	[1.80]
Decile	NYSE	VW	0.36	0.41	0.33	0.34	0.24	0.26
			[3.98]	[4.58]	[3.82]	[3.87]	[2.70]	[2.89]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.22	0.24	0.19	0.19	0.14	0.15
			[2.85]	[3.06]	[2.48]	[2.51]	[1.86]	[1.94]
Quintile	NYSE	EW	0.32	0.41	0.32	0.28	0.18	0.16
			[4.39]	[5.69]	[4.87]	[4.32]	[2.97]	[2.65]
Quintile	Name	VW	0.25	0.27	0.22	0.22	0.17	0.18
			[3.29]	[3.53]	[2.96]	[2.95]	[2.31]	[2.36]
Quintile	Cap	VW	0.18	0.22	0.17	0.17	0.12	0.12
			[2.31]	[2.81]	[2.19]	[2.17]	[1.54]	[1.59]
Decile	NYSE	VW	0.32	0.36	0.28	0.28	0.20	0.21
			[3.51]	[3.92]	[3.19]	[3.25]	[2.31]	[2.45]

Table 3: Conditional sort on size and ELS

This table presents results for conditional double sorts on size and ELS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ELS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ELS and short stocks with low ELS .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		ELS Quintiles					ELS Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.51	0.74	0.82	0.89	1.03	0.55	0.61	0.54	0.48	0.38	0.35
		[1.98]	[2.92]	[3.27]	[3.71]	[4.46]	[5.62]	[6.34]	[5.83]	[5.15]	[4.25]	[3.80]
	(2)	0.64	0.73	0.76	0.88	0.97	0.33	0.40	0.30	0.27	0.24	0.22
		[2.76]	[3.11]	[3.28]	[3.85]	[4.54]	[3.50]	[4.31]	[3.36]	[3.00]	[2.63]	[2.40]
	(3)	0.54	0.57	0.82	0.77	0.96	0.42	0.45	0.36	0.31	0.30	0.26
		[2.66]	[2.66]	[3.72]	[3.76]	[4.91]	[5.04]	[5.39]	[4.58]	[3.87]	[3.74]	[3.22]
	(4)	0.57	0.62	0.76	0.75	0.78	0.22	0.26	0.16	0.16	-0.03	-0.01
		[2.93]	[3.17]	[3.84]	[3.93]	[4.29]	[2.43]	[2.95]	[1.98]	[1.98]	[-0.35]	[-0.10]
	(5)	0.44	0.50	0.45	0.57	0.68	0.24	0.26	0.21	0.23	0.16	0.19
		[2.67]	[2.91]	[2.68]	[3.61]	[4.28]	[2.67]	[2.90]	[2.30]	[2.54]	[1.80]	[2.09]

Panel B: Portfolio average number of firms and market capitalization

		ELS Quintiles Average n					ELS Quintiles Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	308	306	305	305	300	26	26	27	22	21
	(2)	88	87	88	88	87	44	43	44	43	43
	(3)	66	65	66	65	65	78	77	80	81	80
	(4)	56	56	56	57	57	170	173	181	177	183
	(5)	54	55	54	54	54	1220	1367	1568	1491	1765

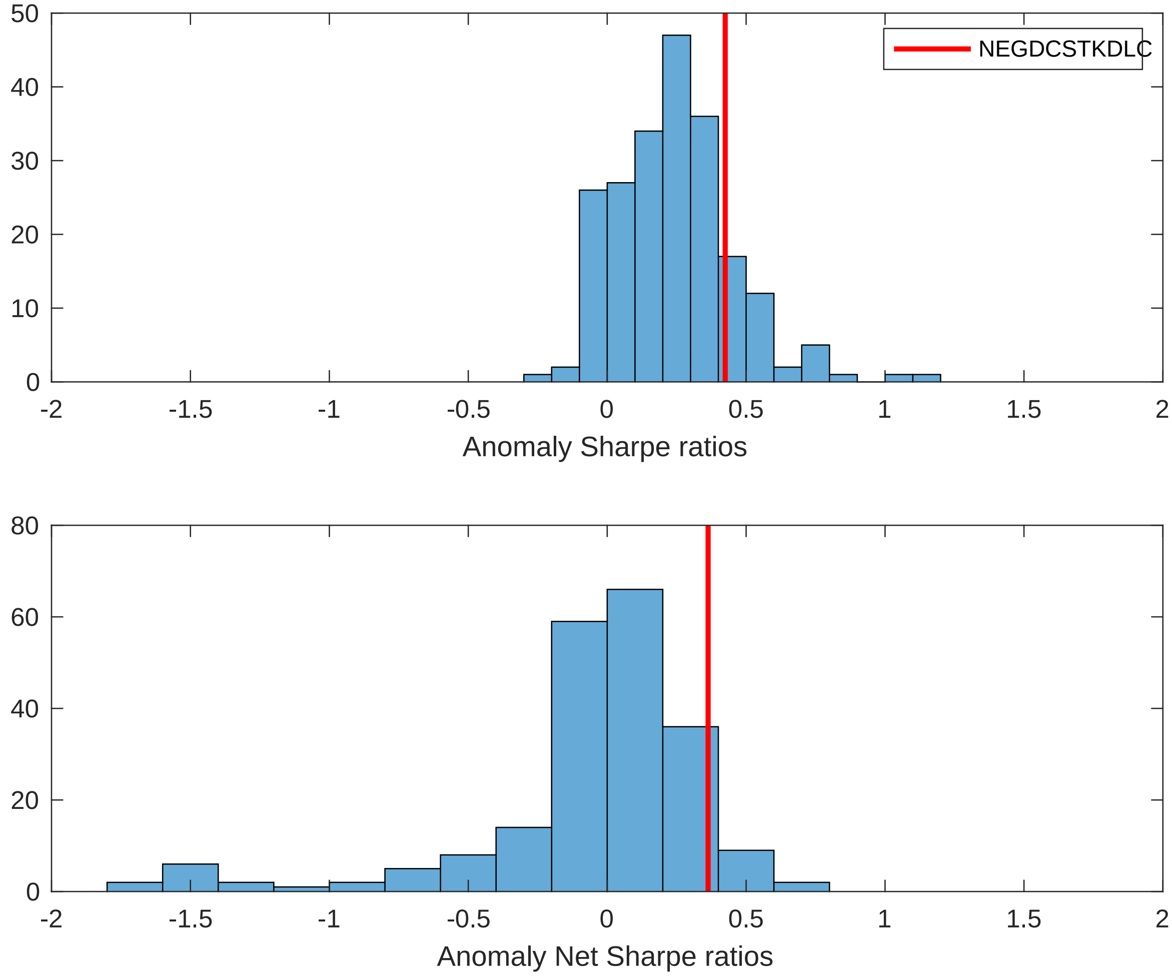


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ELS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

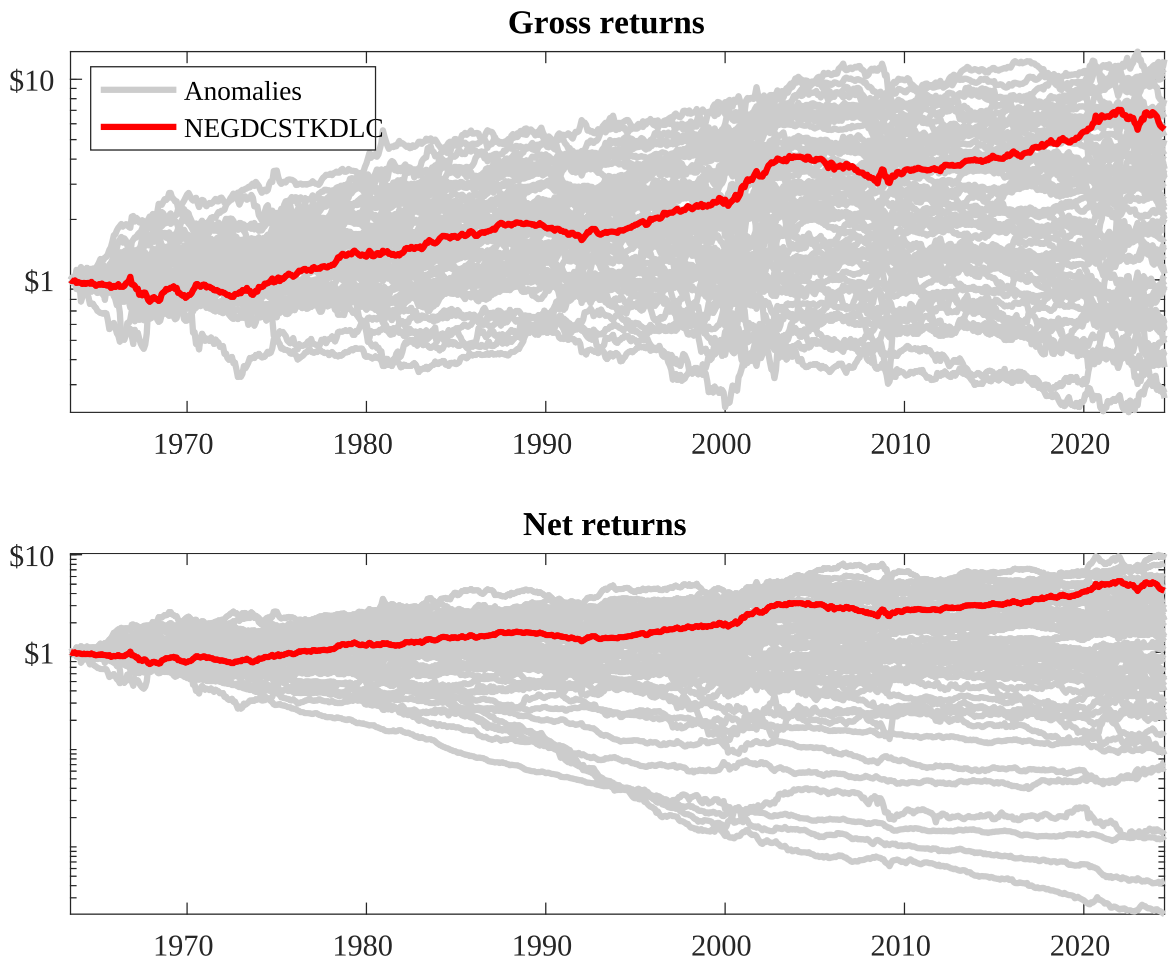
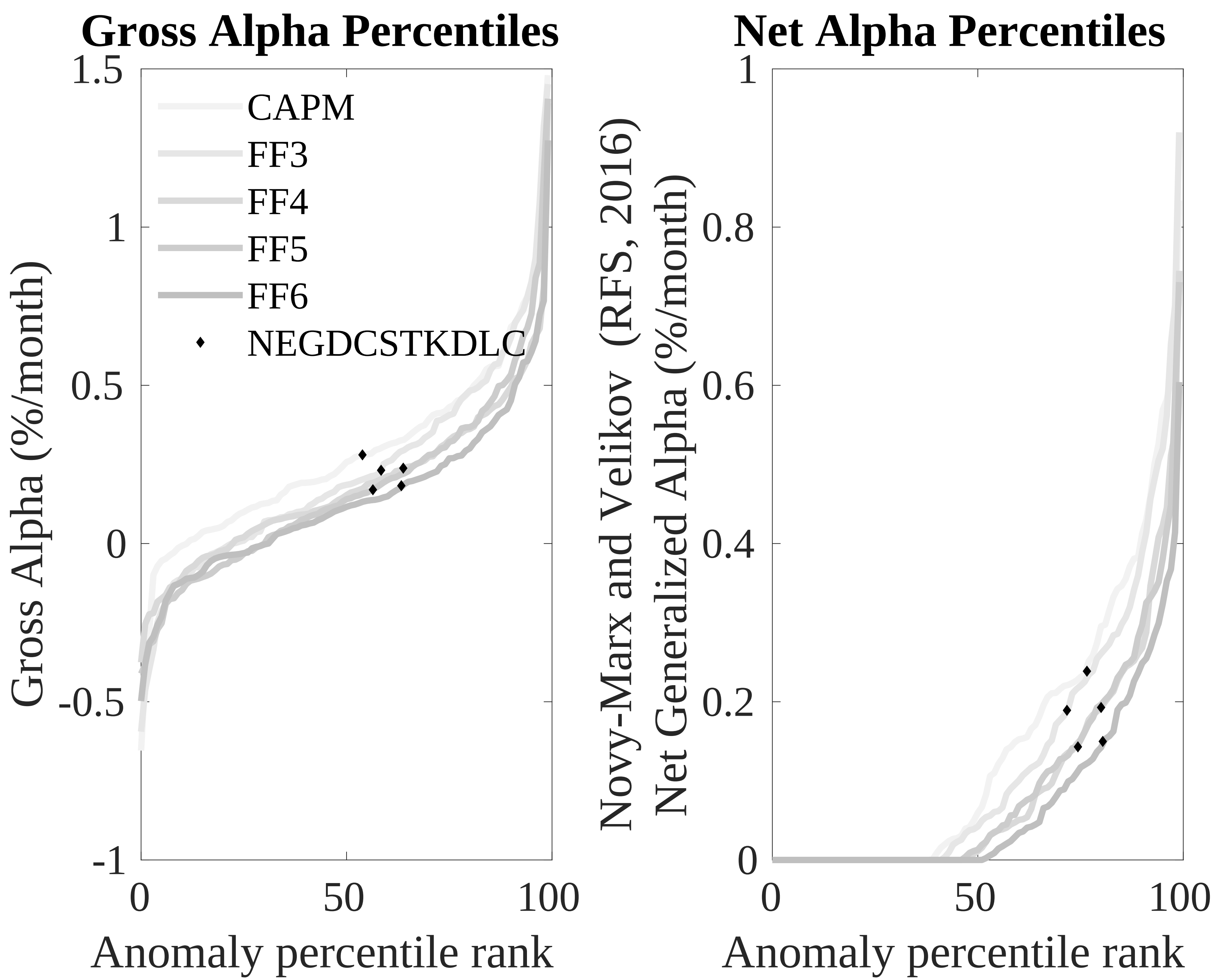


Figure 3: Dollar invested.
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ELS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDCSTKDLC
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100

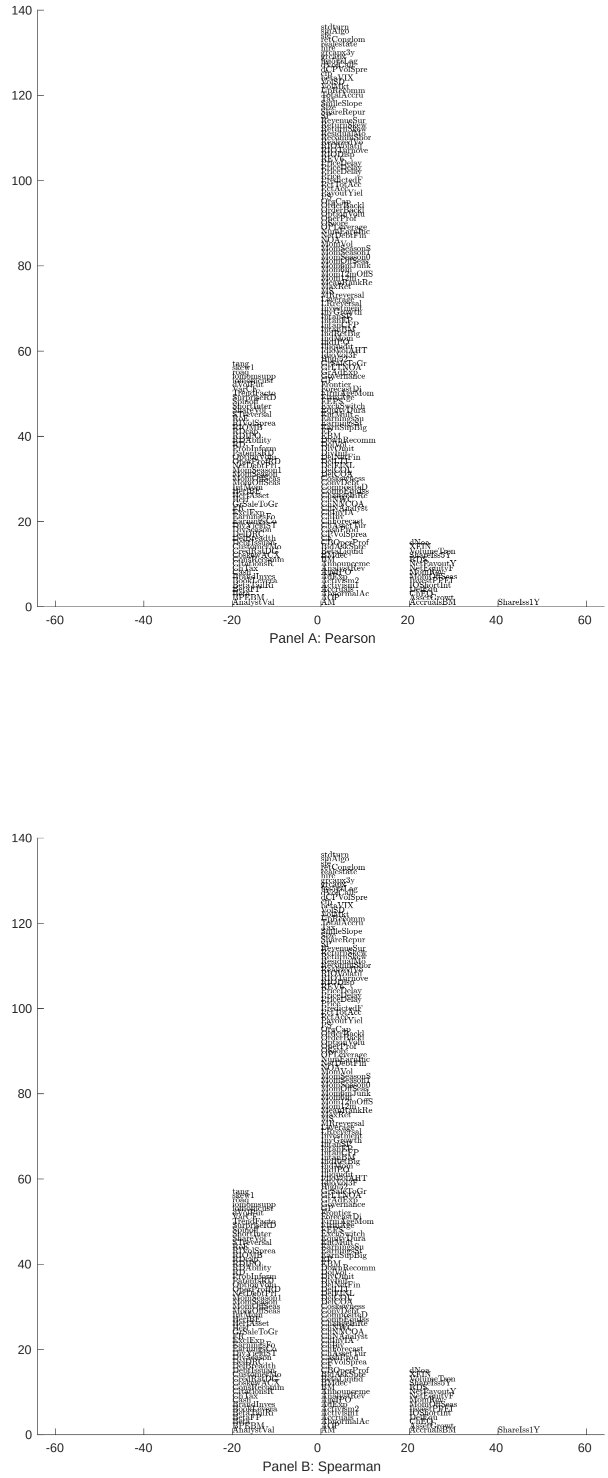


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 209 filtered anomaly signals with ELS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

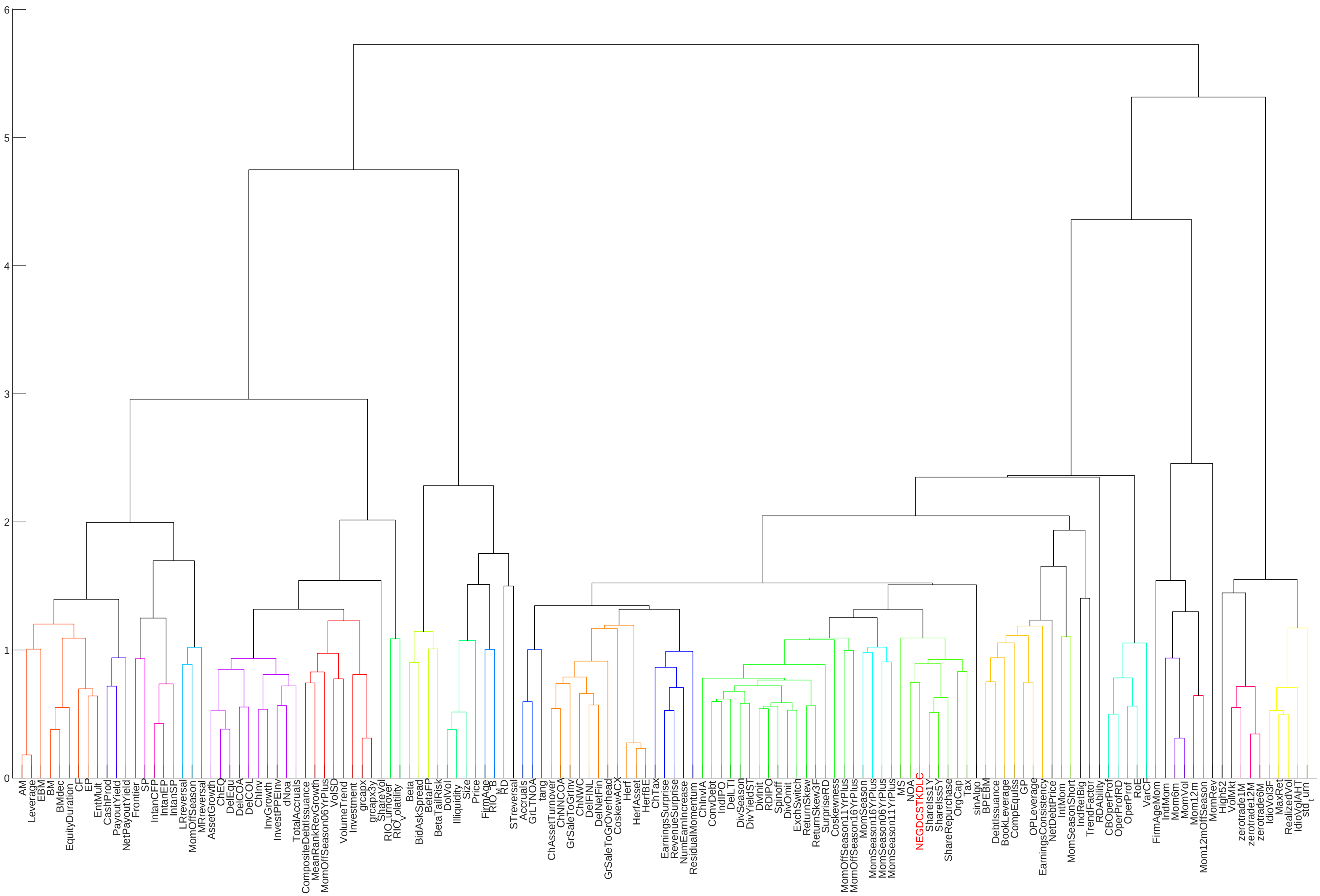


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

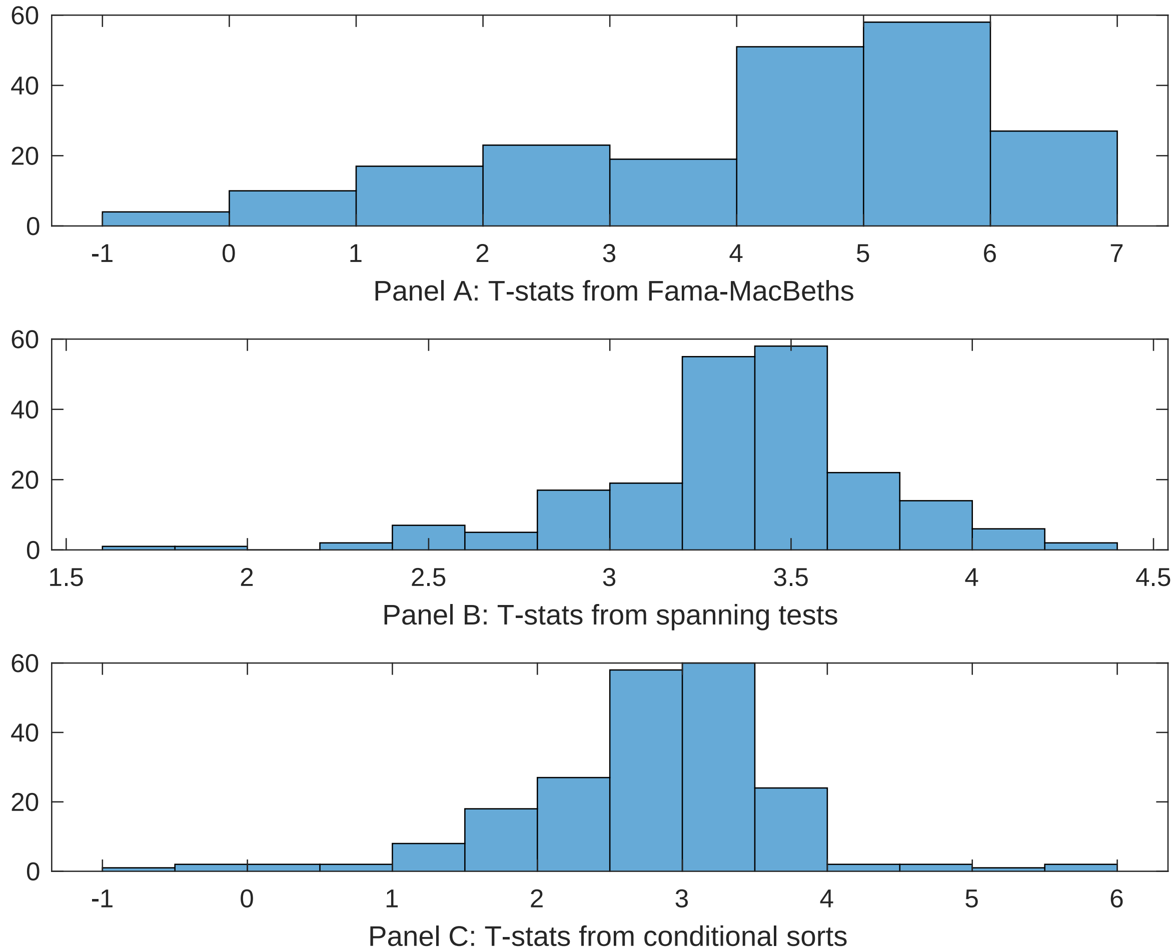


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of ELS conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ELS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ELS} ELS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ELS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ELS. Stocks are finally grouped into five ELS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ELS trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on ELS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ELS} ELS_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13	0.12	0.17	0.13	0.12	0.13	0.13
	[6.01]	[5.34]	[6.93]	[6.44]	[5.67]	[5.31]	[4.91]
ELS	0.68	0.69	0.64	0.70	0.68	0.78	0.42
	[4.82]	[4.60]	[3.88]	[4.70]	[4.19]	[3.51]	[2.51]
Anomaly 1	0.15						0.65
	[6.33]						[2.73]
Anomaly 2		0.20					0.11
		[1.69]					[2.24]
Anomaly 3			0.43				-0.94
			[3.50]				[-0.65]
Anomaly 4				0.30			0.30
				[5.58]			[6.41]
Anomaly 5					0.13		0.12
					[3.26]		[2.02]
Anomaly 6						0.18	-0.14
						[2.91]	[-1.59]
# months	715	704	720	715	720	630	622
$\bar{R}^2(\%)$	0	1	0	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ELS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ELS} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.17	0.21	0.16	0.12	0.18	0.23	0.19
	[2.16]	[2.70]	[2.13]	[1.52]	[2.27]	[2.75]	[2.37]
Anomaly 1	22.60						2.63
	[5.75]						[0.49]
Anomaly 2		17.08					11.95
		[5.68]					[2.92]
Anomaly 3			26.81				30.44
			[6.42]				[4.66]
Anomaly 4				24.86			16.67
				[6.01]			[3.14]
Anomaly 5					12.95		-10.45
					[3.32]		[-1.74]
Anomaly 6						7.33	-12.09
						[1.92]	[-2.51]
mkt	2.57	4.65	2.87	3.12	1.75	2.82	3.55
	[1.39]	[2.44]	[1.55]	[1.69]	[0.93]	[1.36]	[1.74]
smb	-2.46	-0.54	-4.15	-1.68	-3.41	1.87	-3.55
	[-0.92]	[-0.20]	[-1.56]	[-0.63]	[-1.26]	[0.57]	[-1.09]
hml	5.29	-0.83	2.50	7.85	4.09	6.08	2.55
	[1.48]	[-0.22]	[0.70]	[2.20]	[1.12]	[1.61]	[0.66]
rmw	-5.89	-3.62	7.54	-2.57	7.46	7.61	4.21
	[-1.41]	[-0.90]	[2.10]	[-0.66]	[2.03]	[1.69]	[0.86]
cma	6.95	6.67	-4.82	8.96	8.11	6.81	-19.19
	[1.21]	[1.15]	[-0.73]	[1.60]	[1.23]	[1.11]	[-2.64]
umd	-2.67	-0.28	-2.10	-3.26	-1.31	-1.38	-2.20
	[-1.45]	[-0.15]	[-1.15]	[-1.77]	[-0.70]	[-0.71]	[-1.15]
# months	728	728	732	728	732	630	626
$\bar{R}^2$(%)	11	11	12	12	8	6	14

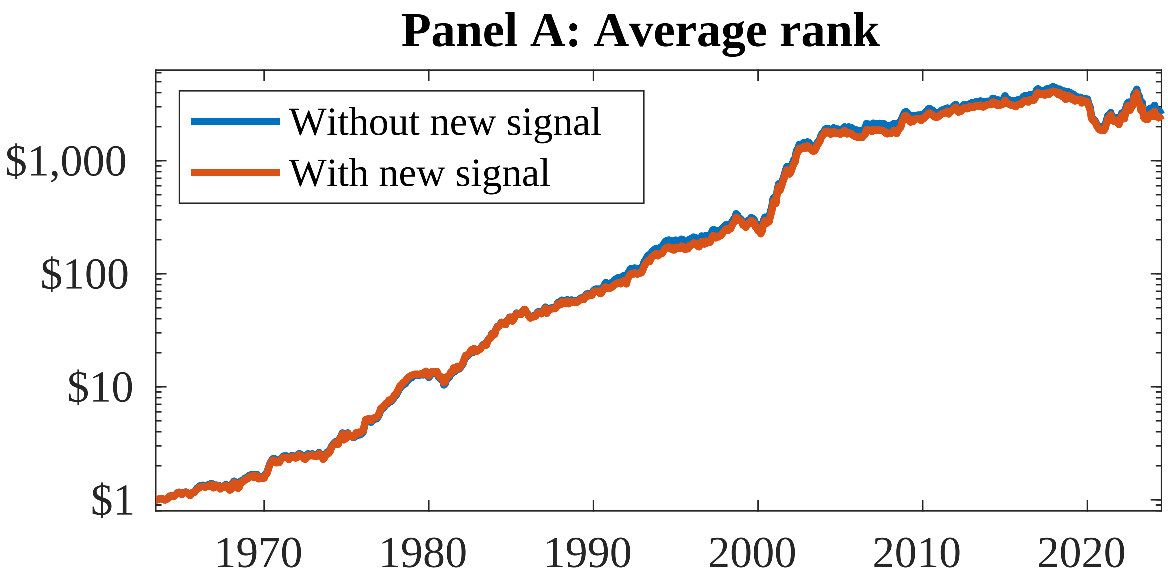


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as ELS. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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